

Curriculum Vitae

Dr. Wooyeon Jo

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Education and Research History

- *July. 2022 – Current* : Postdoctoral Research Associate, Virginia Commonwealth University
PI: Prof. Irfan Ahmed
Major: Digital Forensics and Threat Intelligence
- *Mar. 2015 – Feb. 2022* : Ph.D. in Computer Engineering, Ajou University ICS Lab
(Integrated Ph.D. Program)
PI: Prof. Taeshik Shon
Major: Digital Forensics and Cyber Security
- *Mar. 2011 – Feb. 2015* : B.S. in Computer Engineering, Ajou University
Major: Computer Engineering

Honors

- USCYBERCOM **Guardian** Award by Mike Clark, the Deputy Director of Plans and Policy (J5) of U.S. Cyber Command. The award ceremony was organized at National Cryptologic Museum, Annapolis, MD, February 2025
- USCYBERCOM **Commander** Award and **Guardian** Award (as First Author) by Four-star General Timothy D. Haugh, the Commander of US Department of Defense Cyber Command. The award ceremony was organized at US Naval Academy, Annapolis, MD, April 2024
 - With USCYBERCOM Challenge Coin awarded by Four-star General Timothy D. Haugh.
 - With USCYBERCOM Challenge Coin awarded by Rear Admiral Dennis Velez, the director of plans and policy of United States Cyber Command
- Best Student Paper Award, 23rd Annual Digital Forensics Research Conference (DFRWS US'23), hosted at Johns Hopkins University, Baltimore, MD, July 2023
- Graduate Student Representative on February, 2022
- Student Scholarship from DFRWS USA 2019
- Future Computing Scholarship 2014
- Information Computer Scholarship 2013

Graduated Theses

- Doctor of Philosophy
 - Smart Digital Forensics:
An Application of Supervised Learning to Industrial Network and System
 - Committee: Dr. Manpyo Hong(Chair), Dr. Taeshik Shon, Dr. Jin Kwak. Dr. Tae-Sun Chung(Ajou University), Dr. Changhoon Lee(Seoul National University of Science & Technology)

Research Interests

- Cyber Security
 - Industrial Network Security
 - Industrial Control System, In-Vehicle Network (IVN), IIoT
 - Abnormal Behavior Detection
 - Artificial Neural Network, Deep Learning
- Digital Forensics
 - ICS Forensics
 - Programmable Logic Controller (PLC), Proprietary Protocols
 - Forensicable System Design
 - File System
 - Volatile Memory
 - Smart Devices

Publications

International Journals (SCI/E)

- S Kim, **Wooyeon Jo**, et al. "Two-Phase industrial control system anomaly detection using communication patterns and deep learning." *Electronics* 13.8 (2024): 1520.
- M Kim, Y Shin, **W Jo**, T Shon. "Digital forensic analysis of intelligent and smart IoT devices." *The Journal of Supercomputing* 79.1 (2023): 973-997.
- **Wooyeon Jo**, et al. "Automatic Whitelist Generation System for Ethernet based In-Vehicle Network", *Computers in Industry*. (2022)
- **Wooyeon Jo**, Hyunsoo Chang, and Taeshik Shon. "Digital forensic science approach by file recovery research." *The Journal of Supercomputing* 74.8 (2018): 3704-3725.
- **Wooyeon Jo**, et al. "Digital forensic practices and methodologies for AI speaker ecosystems." *Digital Investigation* 29 (2019): S80-S93.
- **Wooyeon Jo**, et al. "Packet Preprocessing in CNN-Based Network Intrusion Detection System." *Electronics* 9.7 (2020): 1151.
- Kim, Myungjong, **Wooyeon Jo**, Taeshik Shon. "Visualization for internet of things: power system and financial network cases." *Multimedia Tools and Applications* 78.3 (2019): 3241-3265.
- Lee, Seokjun, **Wooyeon Jo**, Taeshik Shon. "ExtSFR: scalable file recovery framework based on an Ext file system." *Multimedia Tools and Applications* (2019): 1-19.
- Shin, Yeonghun , **Wooyeon Jo**, et al. "Certificate Injection-Based Encrypted Traffic Forensics in AI Speaker Ecosystem." *Forensic Science International: Digital Investigation* 33 (2020): 301010.
- Kim, SungJin, **Wooyeon Jo**, and Taeshik Shon. "APAD: autoencoder-based payload anomaly detection for industrial IoE." *Applied Soft Computing* 88 (2020): 106017.
- Kim, HyunJin , **Wooyeon Jo**, et al. "Unknown Payload Anomaly Detection Based on Format and Field Semantics Inference in Cyber-Physical Infrastructure Systems." *IEEE Access* (2021).

International Conference

- **Wooyeon Jo**, and Irfan Ahmed. "Oops, It Halted Again: Exploiting {PLC} Memory for Fun and Profit in Industrial Control Systems." 19th USENIX WOOT Conference on Offensive Technologies (WOOT 25). 2025.
 - *With CVE-2024-11737, CVSS 9.3 CRITICAL*
- Ayub, Adeen, **Wooyeon Jo**, and Irfan Ahmed. "Charlie, Charlie, Charlie on Industrial Control Systems: PLC Control Logic Attacks by Design, Not by Chance." 2024 IEEE International Symposium on Hardware Oriented Security and Trust (HOST). IEEE, 2024.
- **Wooyeon Jo**, et al. "Nazar: Stateful Intrusion Detection System for Control Logic Attacks in Industrial Control Systems". USCYBERCOM's Cyber RECon 2024. USCYBERCOM 2024.
 - *USCYBERCOM's Cyber RECon 2024 Commander's Award – Best in Show, 2024 ([link](#))*
- A Ayub, N Zubair, H Yoo, **W Jo**, I Ahmed. "Gadgets of gadgets in industrial control systems: Return oriented programming attacks on PLCs." 2023 IEEE International Symposium on Hardware Oriented Security and Trust (HOST). IEEE, 2023.
- Qasim, Syed Ali, **Wooyeon Jo**, and Irfan Ahmed. "PREE: Heuristic builder for reverse engineering of network protocols in industrial control systems." Forensic Science International: Digital Investigation 45 (2023): 301565.
 - *(Best Student Paper Award)*
- A Ayub, **W Jo**, SA Qasim, I Ahmed "How are industrial control systems insecure by design? a deeper insight into real-world programmable logic controllers." *IEEE Security & Privacy* 21.4 (2023): 10-19.
- **Wooyeon Jo**, Hyunsoo Chang, and Taeshik Shon. "Digital forensic approach for file recovery in Unix systems: Research of data recovery on Unix file system." 2016 IEEE Information Technology, Networking, Electronic and Automation Control Conference. IEEE, 2016.
- **Wooyeon Jo**, et al. "Digital forensic practices and methodologies for AI speaker ecosystems." DFRWS USA 2019, 2019
- Eo, Soowoong, **Wooyeon Jo**, et al. "A phase of deleted file recovery for digital forensics research in Tizen." 2015 5th International Conference on IT Convergence and Security

(ICITCS). IEEE, 2015.

- Kim, SungJin, **Wooyeon Jo**, and Taeshik Shon. "A novel vulnerability analysis approach to generate fuzzing test case in industrial control systems." 2016 IEEE Information Technology, Networking, Electronic and Automation Control Conference. IEEE, 2016.
- Jeong, Jaehan, **Wooyeon Jo**, et al. "Backtracking and Visualization Techniques for Financial Network Security." 2018 IEEE International Conference of Safety Produce Informatization (IICSPI). IEEE, 2018.
- Kim, Hyunjin, **Wooyeon Jo**, et al. "A novel security framework for industrial iot based on isa 100.11 a." International conference on heterogeneous networking for quality, reliability, security and robustness. Springer, Cham, 2018.
- Kim, Hyungchan, **Wooyeon Jo**, et al. "Digital Forensic Analysis using Android Application Cache Data." 2019 International Conference on Platform Technology and Service (PlatCon). IEEE, 2019.
- Shin, Yeonghun, **Wooyeon Jo**, et al. "An Security Analysis of Ext Filesystem metadata." 2019 4th Technology Innovation Management and Engineering Science International Conference (TIMES-iCON). IEEE, 2019.
- Shin, Yeonghun , **Wooyeon Jo**, et al. "Certificate Injection-Based Encrypted Traffic Forensics in AI Speaker Ecosystem." DFRWS USA 2020, 2020

Patents(Korea)

- [Granted] METHOD AND APPARATUS FOR VISUALIZING ANOMALY DETECTION IN NETWORK FORENSICS, 10-1976395, 2019.05.02, doi:10.8080/1020170022915
- [Granted] METHOD FOR MANAGING DATA OF FILE SYSTEM AND COMPUTING SYSTEM USING THE SAME, 10-1836380, 2018.03.02, doi:10.8080/1020160063614
- [Granted] METHOD FOR MANAGING DATA OF FILE SYSTEM AND COMPUTING SYSTEM USING THE SAME, 10-1769778, 2017.08.14, doi:10.8080/1020160055566
- [Granted] APPARATUS AND METHOD FOR DIGITAL FORENSIC FOR IOT SYSTEM BASED ON CLOUD, 10-2156174, 2020.09.09, doi:10.8080/1020190075441

Programming Skills

- Expert in Python/C/JAVA programming with network-related project experiences.
- Experienced in programming with Web with network security visualization project experiences.
- Experienced in programming with C++ on Windows, Linux and Android NDK environments.

Teaching Experience

As a graduate student in the doctoral program, I took one or more TAs every semester. The following describes a case where I gained teaching experience by taking lab classes for the entire semester during the course. I have provided these hands-on lectures to not only undergraduate students of Ajou University, but also military personnel and gifted education.

- **Guest Lecturer / Interim Instructor, Memory Forensics**
 - Spring 2023 in Virginia Commonwealth University, ICS Security and Attacks
 - Spring 2024 in Virginia Commonwealth University, ICS Security and Attacks
 - Fall 2025 in Virginia Commonwealth University, Memory forensics with Volatility 3
- **Teaching Assistant, Software Application Security**
 - Fall 2017 in Ajou University as TA, Software application security
 - Fall 2018 in Ajou University as TA, Software application security
 - Led several lectures on Digital Forensics (file system, memory, automotive, mobile, and smart devices), Network Security, and ICS on behalf of the professor

Projects

- File system forensic analysis tool development, Supreme Prosecutors' Office Republic of Korea, 2012.10.01 ~ 2017.08.31.
- BK21 Plus, National Research Foundation of Korea, 2013.09 ~ 2020.08
- SG heterogeneous protocol gateway security module development, KEPCO(Korea Electric Power Corporation) KDN (Knowledge, Data and Network), 2014.12 ~ 2015.05
- A Study on Vulnerability Analysis Technology in Implementation of Control Protocol, National Security Research Institute, 2015.04 ~ 2015.11
- Development of digital evidence analysis and restoration software based on open platform file system in IoT environment, Ajou University LINK Project Team, 2015.07 ~ 2016.01
- Control system network forensic architecture study, National Security Research Institute, 2016.4. ~ 2016.10.
- Industrial IoT target vulnerability fuzzing and digital forensic technology development, Ajou University LINK Project Team, 2016.07. ~ 2017.01.
- ISA 100.11a based power control system wireless security technology development, KEPCO(Korea Electric Power Corporation) Research Institute, 2016.12 ~ 2017.12.
- Industrial wireless communication protocol security threat analysis study, National Security Research Institute, 2018.04 ~ 2018.10
- Preemptive recognition of manufacturing process abnormalities to solve the problem of smart factory shutdown due to cyber attacks, Institute for Information & Communication Technology Planning & Evaluation (IITP), 2018.04 ~ 2020.12
- Development of digital forensic integrated platform, Institute for Information & Communication Technology Planning & Evaluation (IITP), 2018.05 ~ 2020.12
- A study on the storage data system of cloud server-based artificial intelligence assistants and devices, Supreme Prosecutors' Office Republic of Korea, 2018.06 ~ 2018.12
- Advancement of power control system abnormal behavior detection algorithm and research on detection equipment development, KPX(Korea Power Exchange), 2018.07 ~ 2020.07
- AI Speaker Forensic Analysis, Korea Institute of Information Security & Cryptology, 2019.03~2019.11
- Deep Learning-based Abnormal Behavior Detection Technology Research in Vehicle Ethernet Environment, HYUNDAI Research Project for Future Technology, 2019.08~2020.02
- A study on how to completely delete data in Android OS (9,10), National Security Research Institute, 2020.04 ~ 2020.10
- A study on analysis techniques for wearable devices (smart watches and bands). Korean National Police Agency, 2020.09 ~ 2020.12

- Reliable intelligent threat detection engine research for national infrastructure cybersecurity, National Security Research Institute, 2018.06 ~ 2023.02
- Vehicle Infotainment Forensic Analysis, Korea Institute of Information Security & Cryptology, 2020.03 ~ 2021.11
- Ethernet Communication Diagnostics, HYUNDAI, 2021.06 ~ 2022.06
- Funded by the U.S. Department of Homeland Security under Grant Award Number 17STCIN00001-05-00, 2022.09 ~ Current

Professional References

- Dr. Irfan Ahmed, Virginia Commonwealth University, iahmed3@vcu.edu
- Dr. Taeshik Shon, Ajou University, tsshon@ajou.ac.kr
- Dr. Adeen Ayub, ayubax@jmu.edu
- Dr. Muhammad Haris Rais, Virginia State University, mrais@vsu.edu
- Dr. Syed Ali Qasim, Grand Valley State University, qasims@gvsu.edu

Personal References available upon request

I hereby declare that the above information is true and correct in every respect to the best of my knowledge.

09/2025

Wooyeon Jo